

Physical-Time Action Relaxation in an Elastic Collision Bath

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Abstract

An elastic repeated-collision model produces an action-valued covariance observable that relaxes exponentially in physical time to a positive constant. We derive its rate and plateau directly from the masses, bath velocity variance and collision frequency. Bounded bath velocities keep every collision below a fixed speed ceiling. The exact parameter dependence isolates the additional law needed for a universal action scale: energy and momentum conservation determine the collision map, while the reservoir statistics set the plateau. An equal-mass two-speed gas supplies a spatial realization: independent Poisson spacings generate reversal rate ρu and plateau mu/ρ . Ordered streams instead give periodic tagged motion and a zero plateau at the same mass, density, speed and mean collision frequency.

1 A momentum receiver and a classical stochastic clock

A tracer of mass $m \geq M > 0$ meets fresh bath particles of mass M on the line. Collisions occur at an independent Poisson rate $\nu > 0$. Incoming bath velocities W_n are independent, identically distributed, centered and obey $|W_n| \leq w < c/3$, with variance $s^2 > 0$. Initial tracer velocity is independent of the bath and clock, centered, and bounded by w . Between collisions it is constant; $X(t) = X(0) + \int_0^t V(r) dr$.

The nontrivial one-dimensional elastic collision map is

$$V' = aV + bW, \quad W' = \frac{2m}{m+M}V - aW, \quad a = \frac{m-M}{m+M}, \quad b = \frac{2M}{m+M} = 1-a. \quad (1)$$

Direct substitution gives $mV' + MW' = mV + MW$ and $m(V')^2 + M(W')^2 = mV^2 + MW^2$. Since $0 \leq a < 1$, the tracer update is a convex combination and preserves $|V| \leq w$. The departing particle obeys $|W'| \leq (3m-M)w/(m+M) < 3w < c$.

The model specifies a refreshed reservoir and a velocity-independent clock. These are Barkai's collision-model premises [3, pp. 3–5, (2)–(12)], specialized here to a bounded bath and $m \geq M$. Each collision accounts for momentum and energy transfer; the complete stream supplies fresh particles and removes departing ones. A spatial gas realization would additionally derive encounter rates and incoming velocity statistics. The speed ceiling here is a kinematic restriction on Newtonian collisions in the reservoir frame, rather than a Lorentz-covariant collision law.

2 Stationary covariance and the action plateau

Define the two inverse-time rates

$$\gamma = \nu(1-a) = \frac{2\nu M}{m+M}, \quad \beta = \nu(1-a^2) = \frac{4\nu m M}{(m+M)^2}. \quad (2)$$

The velocity generator acts on bounded functions as $Lf(v) = \nu\{\mathbb{E}[f(av + bW)] - f(v)\}$.

Proposition 1 (Collision plateau). *The stationary velocity law is unique on $[-w, w]$ and is represented by the convergent series $V_* \stackrel{d}{=} b \sum_{j=0}^{\infty} a^j W_j$. Its variance and autocovariance are*

$$S_* := \text{Var } V_* = \frac{b^2 s^2}{1 - a^2} = \frac{M}{m} s^2, \quad C(r) = S_* e^{-\gamma r}. \quad (3)$$

Thus the stationary displacement observable $h(\Delta) = m \text{Var}[X(t + \Delta) - X(t)]/\Delta$ satisfies

$$h(\Delta) = H_* \left[1 - \frac{1 - e^{-\gamma \Delta}}{\gamma \Delta} \right], \quad H_* := \frac{2mS_*}{\gamma} = \frac{(m + M)s^2}{\nu} > 0. \quad (4)$$

Proof. The series converges absolutely and stays in $[-w, w]$, since $b \sum a^j = 1$ (for $a = 0$ it is simply W_0). Separating its first term proves invariance under one collision. Coupling two initial velocities to the same clock and bath multiplies their difference by a^{N_t} . Its mean contraction factor is $\mathbb{E}a^{N_t} = e^{-\nu(1-a)t}$, also for $a = 0$ with $0^0 = 1$. Hence invariant probability laws on the bounded interval coincide. Independence and centering give the variance in (3). Conditioning on the collision count yields $\mathbb{E}[V(t + r) \mid V(t)] = e^{-\gamma r} V(t)$; stationary covariance follows. Integrating the covariance over the displacement square gives $h(\Delta) = 2m \int_0^\Delta (1 - r/\Delta) C(r) dr$, which evaluates to (4). $\square$

This proof uses a contracting affine collision update; it requires neither a finite velocity state space nor detailed balance of the refreshed bath. The energy-valued reservoir parameter $\Theta := Ms^2$ gives $mS_* = \Theta$ and $H_* = \Theta(m + M)/(\nu M)$. For a general bounded bath, Θ denotes twice the mean incoming kinetic energy, without assuming a thermodynamic equilibrium distribution. The stationary covariance and Green–Kubo integral also follow from Barbier and Trizac’s field-free Maxwell-kernel specialization [2, p. 19, (89)–(98)]. Their kernel exponent called ν is zero there; our ν is a positive collision frequency.

3 An observable relaxing in physical time

The integrated future-lag covariance defines a second operational quantity:

$$a(t) := 2m \int_0^\infty \text{Cov}[V(t + r), V(t)] dr. \quad (5)$$

It has action units and uses physical preparation time t , with the lag integrated out. Its definition differs from the finite-duration displacement observable; both share the same stationary long-duration plateau.

Proposition 2 (Physical-time relaxation). *For the centered initial ensemble specified above, with $S_0 = \text{Var } V(0)$,*

$$S(t) = S_* + (S_0 - S_*)e^{-\beta t}, \quad (6)$$

$$a(t) = \frac{2mS(t)}{\gamma} = H_* + (a(0) - H_*)e^{-\beta t}, \quad \dot{a} = \beta(H_* - a). \quad (7)$$

In particular a tracer prepared at rest has $a(t) = H_(1 - e^{-\beta t}) > 0$ for every $t > 0$ and $a(t) \rightarrow H_* > 0$ as $t \rightarrow \infty$.*

Proof. The generator gives $Lv = -\gamma v$ and $L(v^2) = -\beta v^2 + \nu b^2 s^2$. Centering is preserved, so $\dot{S} = -\beta S + \nu b^2 s^2 = \beta(S_* - S)$. The conditional mean from Proposition 1 gives, even away from stationarity, $\text{Cov}[V(t + r), V(t)] = e^{-\gamma r} S(t)$. Integration proves (7). $\square$

The original displacement definition retains a distinct duration parameter:

$$h_{\Delta}(t) = \frac{2m}{\gamma\Delta} \int_0^{\Delta} S(t+r)[1 - e^{-\gamma(\Delta-r)}] dr. \quad (8)$$

Indeed, for $r \leq s$ the covariance inside its double integral is $e^{-\gamma(s-r)}S(t+r)$; integrating over s proves the formula. At fixed Δ , it approaches the stationary value in (4) exponentially at rate β . At every fixed t it tends to zero as $\Delta \downarrow 0$, with $h_{\Delta}(t) \sim mS(t)\Delta$ when $S(t) > 0$. Its long-duration limit is H_* , since the transient contribution to (8) is bounded in magnitude by $2m|S_0 - S_*|e^{-\beta t}/(\gamma\beta\Delta)$.

4 What would select a common constant?

For fixed bath mass, variance and clock, the plateau grows as $m + M$. Achieving the same H_* for different tracer masses requires

$$\nu_m = \frac{(m + M)s_m^2}{H_*}. \quad (9)$$

This equation is a diagnostic of a proposed encounter law. Binary energy and momentum conservation alone place no restriction of this form on the external clock or reservoir preparation.

More explicitly, take a fixed admissible bath and a tracer initially at rest. For any $0 < \epsilon \leq 1$, replace every incoming velocity by $W_n^{(\epsilon)} = \epsilon W_n$, keeping masses and clock unchanged. Every elastic collision, centering assumption and speed bound remains valid. Both H_* and $a(t)$ are multiplied by ϵ^2 . Consequently this entire class admits arbitrarily small positive plateaus even though every nondegenerate member relaxes to its own positive constant. This is a countermodel test for these specified stochastic collision premises.

The next model should replace the supplied clock by a spatial encounter law and test how it scales with reservoir velocity and density. The physical-time relaxation mechanism is already explicit: persistent injection of velocity variance balances collisional contraction. A universal quantum action scale requires an additional physical relation that fixes this balance across systems.

Ben-Naim and Krapivsky provide a useful clock comparison: exponential decay in collision number becomes algebraic decay in physical time in their cooling model [4, p. 6, (14)–(17)].

5 A spatial receiver with an exact collision clock

An infinite equal-mass hard-point gas realizes the two-speed action plateau with every particle's speed bounded by $u < c$. Its encounter clock follows from the initial spatial statistics. This is the dichotomic Jepsen-gas case analyzed by Balakrishnan, Bena and Van den Broeck [1, pp. 7–8, (11)–(12), pp. 17–19]. The following construction makes its clock explicit.

Fix $m, \rho, u > 0$. Put a tagged particle at the origin and independently put Poisson particles of intensity ρ on the rest of the line. Give every particle an independent sign, with velocities $+u$ and $-u$ equally likely. This is the Palm preparation: a typical particle is placed at zero, with independent gas on either side. Particles move freely between instantaneous elastic collisions, exchanging velocities and preserving their spatial order. Each collision conserves momentum and kinetic energy. The initial energy per unit length has mean $\rho mu^2/2$; the reservoir has infinite extent.

The equivalent free-trajectory construction lets particles cross and exchanges their physical labels at crossings. Each marked Poisson stream is translated rigidly, so the untagged

homogeneous marked-gas law is invariant. A tagged path on $[0, T]$ lies inside $[-uT, uT]$; only free lines starting in $[-2uT, 2uT]$ can intersect that region during this interval. There are almost surely finitely many such lines and crossings. This constructs the local dynamics without an accumulation of collisions. Positions are continuous and piecewise linear; velocities are right-continuous with impulsive changes.

Proposition 3 (Spatially generated reversals). *In this preparation the tagged velocity is a stationary symmetric two-state Markov chain with reversal rate $\lambda = \rho u$. Consequently*

$$C(s) = u^2 e^{-2\rho u s}, \quad (10)$$

$$h(\Delta) = \frac{mu}{\rho} \left[1 - \frac{1 - e^{-2\rho u \Delta}}{2\rho u \Delta} \right], \quad H_* = \frac{mu}{\rho} > 0. \quad (11)$$

Here $h(\Delta) = m \text{Var}[X(t+\Delta) - X(t)]/\Delta$, and the limit is $\Delta \rightarrow \infty$ at fixed m, ρ, u in the infinite gas.

Proof. Condition first on tagged initial velocity $+u$. Write $B_1 < B_2 < \dots$ for the initial positions of negative-velocity free lines to its right, and $-A_1 > -A_2 > \dots$ for positive-velocity lines to its left; set $A_0 = B_0 = 0$. Independent thinning makes (A_j) and (B_j) independent Poisson arrival sequences of intensity $\rho/2$. Their successive gaps are independent exponentials of that rate.

The tag first switches from the line ut to $B_1 - ut$, then to $-A_1 + ut$, then to $B_2 - ut$, and so on. Lines with the same velocity remain parallel; at each switch the next opposing line is the next unused member of the corresponding ordered sequence. Solving these intersections gives

$$t_{2j-1} = \frac{B_j + A_{j-1}}{2u}, \quad t_{2j} = \frac{B_j + A_j}{2u}.$$

Thus successive waiting times alternate between $(B_j - B_{j-1})/(2u)$ and $(A_j - A_{j-1})/(2u)$: all are independent exponentials of rate ρu . Reflection proves the same law for initial velocity $-u$. The independent uniform initial sign is the stationary law of this chain. Its covariance follows by taking the parity of its Poisson reversal count. Integrating $2m \int_0^\Delta (1 - s/\Delta) C(s) ds$ proves the action formula. $\square$

The mechanical statement agrees with the source's covariance (12). Its discussion distinguishes a tag initially at $\pm u$ from a tag initially at rest: p. 19 gives the ballistic term $e^{-\rho u t} \delta(X \mp ut)$ in the former case, and a Bessel-weighted atom at zero in the latter. Our proposition uses the former preparation throughout.

5.1 The scale supplied by spatial preparation

The mean collision duration is $1/(\rho u)$, and the plateau has the action units of momentum times mean spacing: $H_* = mu/\rho$. At fixed density, cooling $u \mapsto \epsilon u$ changes the collision rate by ϵ and the plateau by ϵ , rather than the quadratic scaling of the independently clocked bath. At fixed u , increasing ρ also closes the plateau. Each family retains the hard speed ceiling. A shared value K across these gases therefore requires $\rho_m = mu_m/K$; a preparation law would have to select this relation.

This realization advances the mechanical-clock obligation and inherits the two-speed bridge and cut results through equality of the tagged path laws. The positive plateau concerns long observation windows; microscopic refinement still gives $h(\Delta) \sim mu^2 \Delta \rightarrow 0$. The next test holds m, u, ρ fixed while changing spacing correlations, to identify how much of the plateau is supplied by Poisson preparation. This is a classical impulsive gas in a chosen frame, with a kinematic speed ceiling; a relativistic collision theory or a smooth-force realization would add further dynamics.

6 Ordered streams at fixed mechanical scales

Spatial correlations can cancel displacement while collisions continue at the same mean frequency. Fix the preceding m, ρ, u and put $a = 2/\rho$. Positive free lines start at $a\mathbb{Z} + U_+$ and negative lines at $a\mathbb{Z} + U_-$, with independent uniform phases on $[0, a)$. Their union has density ρ , equal one-particle probabilities for $\pm u$ and energy density $\rho mu^2/2$. Uniform phases make the marked law invariant under translation and under $(U_+, U_-) \mapsto (U_+ + ut, U_- - ut) \pmod{a}$. Elastic label exchange gives the same hard-point dynamics. This changes joint position–velocity correlations, including independence of the marks.

Proposition 4 (Periodic cancellation). *For a typical particle placed at zero, set $d = a/(2u) = 1/(\rho u)$. Its stationary velocity is a uniform-phase square wave with period $2d$. For $\Delta > 0$, put*

$$\delta = \Delta - 2d \left\lfloor \frac{\Delta}{2d} \right\rfloor, \quad s = \min\{\delta, 2d - \delta\}.$$

Then

$$h(\Delta) = \frac{mu^2}{\Delta} \left(s^2 - \frac{2s^3}{3d} \right), \quad 0 \leq h(\Delta) \leq \frac{m}{3\rho^2\Delta} \longrightarrow 0 \quad (\Delta \rightarrow \infty).$$

Proof. The equal stream intensities give the Palm tag a uniform sign. Conditional on positive sign, its own stream is $a\mathbb{Z}$ and the opposing stream has independent uniform phase $U \in (0, a)$. The intersection formulas in Proposition 3 apply with $A_j = ja$ and $B_j = U + (j-1)a$. The first collision time is $R = U/(2u)$, uniform on $(0, d)$; all subsequent waits are d . Reflection treats negative sign. Uniform sign and residual duration give the stationary square-wave law. Its integral has zero displacement over each period $2d$.

For $0 \leq s \leq d$ there is at most one reversal. No reversal has probability $1 - s/d$; a residual time $r \in (0, s)$ gives displacement of magnitude $u|2r - s|$. Sign symmetry gives zero mean, hence

$$\text{Var}[X(t+s) - X(t)] = u^2 \left[(1 - s/d)s^2 + \frac{1}{d} \int_0^s (2r - s)^2 dr \right] = u^2 \left(s^2 - \frac{2s^3}{3d} \right).$$

Periodicity reduces any duration modulo $2d$. For $\delta > d$, zero full-period displacement and stationary phase identify its variance with that of the complementary duration $2d - \delta$. The polynomial increases on $[0, d]$ to $d^2/3$. Multiplication by m/Δ proves the bound. $\square$

Both preparations have collision frequency $1/d = \rho u$, velocity variance u^2 and the same energy density. The ordered case reverses forever while displacement cancels periodically. Its phase law retains conserved global information and is not mixing. Local finiteness ensures existence on every finite window; simultaneous collisions at distinct lattice sites are disjoint binary events, and initial phase coincidences have probability zero.

This comparison isolates correlation control as a further ingredient for preparation-independent positivity. The next test replaces each lattice by an independent stationary renewal stream with the same mean gap, and asks how gap variance controls the plateau. It will distinguish positivity for one preparation from a uniform lower bound over preparations. The explicit ordered construction is derived here; B26 records its bounded prior-art search separately.

7 Return to the partition question

The project's central question concerns a sequence of time partitions $\pi_N = \{0 = t_0^{(N)} < \dots < t_N^{(N)} = T\}$ with mesh tending to zero. The present relaxation law describes preparation

time under a supplied bath. It therefore serves as a comparison mechanism for the partition problem: one must derive, from admissible cut-point refinements, both the quantity that survives and the condition selecting its positive value. The collision calculation isolates how a positive constant can instead enter through reservoir energy and a clock. The next partition test tracks these inputs explicitly when a refinement law is proposed.

References

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